BRENDA

The core database for enzyme data

Antje Jäde, Julia Hauenstein, Lisa Jeske, Mathias Krull, Julia Koblitz, Lorenz C. Reimer & Boyke Bunk

Leibniz Institute DSMZ – German Collection of Microorganisms and Cell Cultures

GLOBAL CORE BIODATA RESOURCE

elixir Core Data Resource

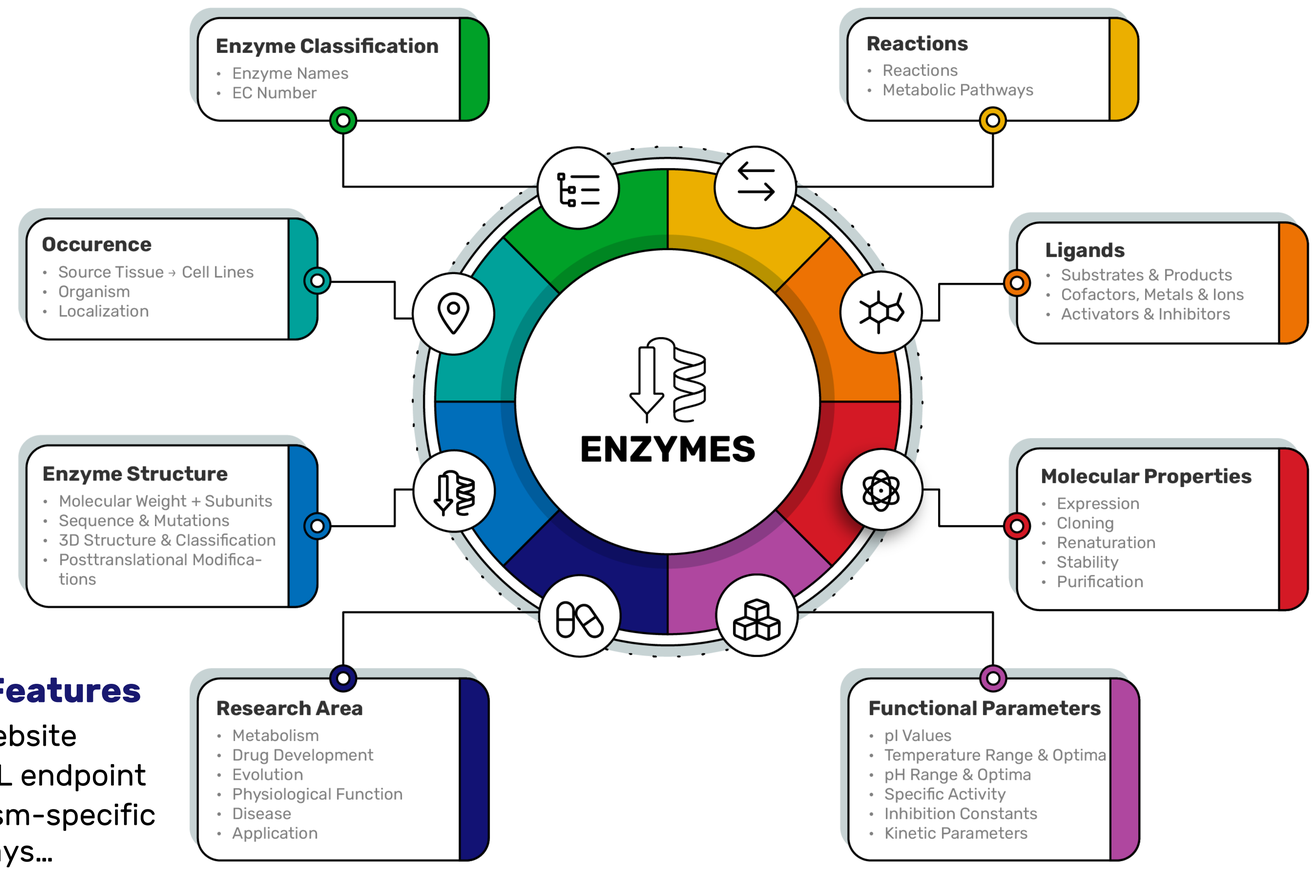
BRENDA Enzyme Database

Over the past 38 years, the BRENDA enzyme database (https://www.brenda-enzymes.org/) has become the most comprehensive resource for manually curated enzyme and ligand-related information in the life sciences.
As part of the German ELIXIR Node - de.NBI, BRENDA has been developed in close collaboration with other biodata resources since 2015. Its significance to the life science community has been highlighted by its designation as an ELIXIR Core Data Resource (since 2018) and a Global Core Biodata Resource (since 2022).

BRENDA at the DSMZ

With the beginning of 2023 BRENDA was completely migrated into the DSMZ – German Collection of Microorganisms and Cell Cultures GmbH with all servers. BRENDA is interconnected with other biodata resources at DSMZ, and its data has been integrated into the Digital Diversity Hub (https://hub.dsmz.de/) as a result of this close collaboration.

Information in BRENDA



Future Features

- New website
- SPARQL endpoint
- Organism-specific pathways…

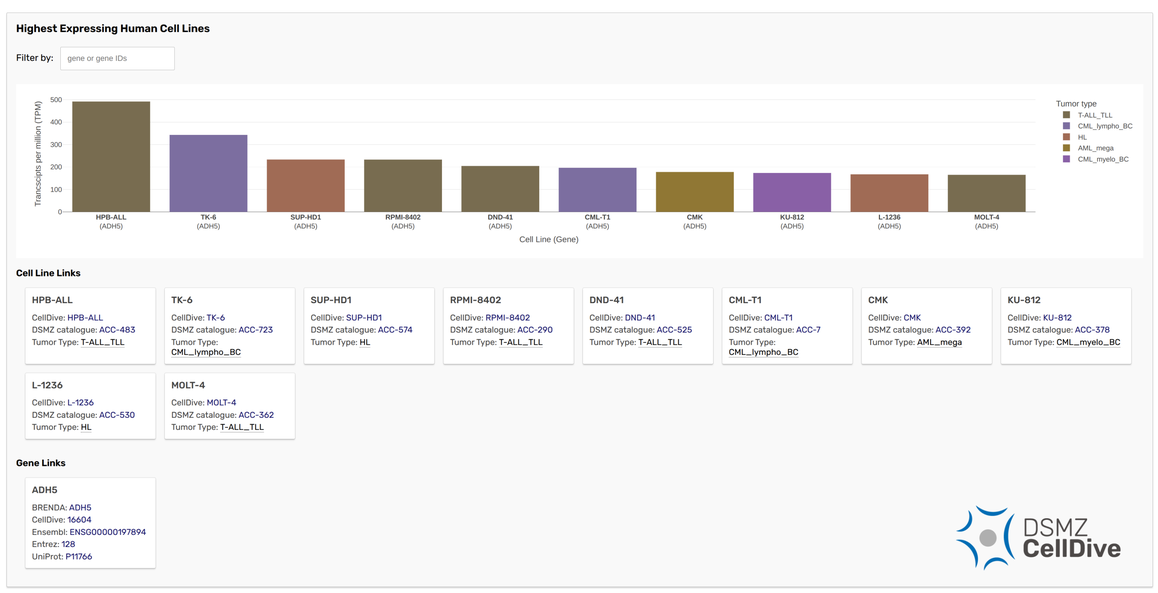
Gene Search and TOP 10 Cell Lines

A new gene search function allows to find enzymes by their names. A colored bar chart displays those ten human cell lines with the highest expression. This tool should help researchers quickly find suitable cell lines to study an enzyme of interest with in vitro models.

New!

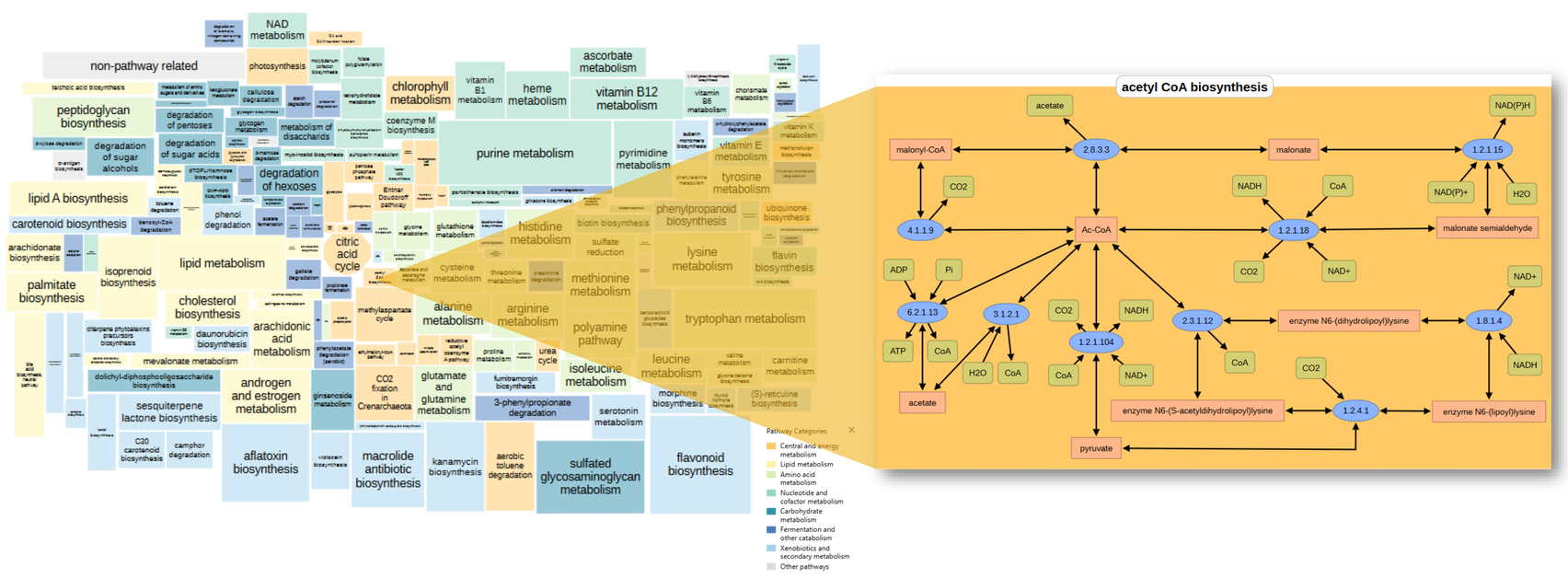
Results **1** - **10** of **2142**

EC Number	Recommended Name Organism	Gene	Ensembl	Entrez	HGNC	UniProt	CellDive Gene ID	Links
2.7.4.6	nucleoside-diphosphate kinase *Homo sapiens*	CMPK1 synonym: CMK	ENSG00000162368	-	HGNC:18170	Q5T0D2	10543	Expessing Cell Lines
2.7.4.6	nucleoside-diphosphate kinase *Homo sapiens*	CMPK1 synonym: CMK	ENSG00000162368	51727	HGNC:18170	A0A494BXC7	10543	Expessing Cell Lines
2.7.4.25	(d)CMP kinase *Vibrio tarriae*	DLR65_04220 synonym: cmk	-	69719448	-	A0A366A8M6	-	-
2.7.4.25	(d)CMP kinase *Bacillus sp*	EGS86_07475 synonym: cmk	-	75084808	-	A0A427RLW5	-	-
2.7.4.25	(d)CMP kinase *Bacillus licheniformis*	CHCC16736_2596 synonym: cmk	-	66215571	-	A0A1Y0XK01	-	-
2.7.4.25	(d)CMP kinase *Bacteroides sp. HMSC067B03*	HMPREF2794_13335 synonym: cmk	-	69592029	-	A0A1E9BY37	-	-
2.7.4.25	(d)CMP kinase *Lactiplantibacillus pentosus*	KSF55_08685 synonym: cmk	-	49394108	-	A0A241RTE9	-	-
2.7.4.25	(d)CMP kinase *Raoultella planticola*	cmk	-	72406201	-	A0A2X2DY52	-	-
2.7.4.25	(d)CMP kinase *Oligella urethralis DNF00040*	HMPREF2130_08665 synonym: cmk	-	61261581	-	A0A095Z4D5	-	-
2.7.4.25	(d)CMP kinase *Methanococcus maripaludis X1*	cmk	-	10982094	-	G0H4B6	-	-



BRENDA Metabolic Pathways

Manually created, global interactive map of metabolic pathways. Zoom in to view details for individual details.

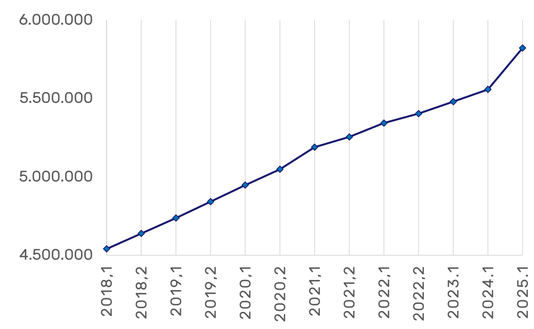


Release 2025

Content in Numbers

EC Classes	8,699
Proteins	253,740
Organisms	15,335
Ligands	278,840
References	173,164

Total Data Points in BRENDA



DSMZ Digital Diversity

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https://brenda-enzymes.org

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contact@brenda-enzymes.org
brenda-enzymes.org
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